In [1]:
```
import pandas as pd
import numpy as np

from scipy.optimize import minimize

pd.set_option("display.max_columns", None)
pd.set_option("display.width", 200)

# ==========================
# 1. 读取清洗后的底表
# 这里必须有历史收益数据，MVO 要用它算协方差矩阵
# ==========================
df_model = pd.read_csv("cleaned_model_data.csv")
df_model["MthCalDt"] = pd.to_datetime(df_model["MthCalDt"])

# ==========================
# 2. 读取 post-2022 的 RF 选股结果
# 这是你前面已经保存好的文件
# ==========================
rf_stock_selection_df = pd.read_csv("rf_top6_stock_selection_post2022.csv
rf_stock_selection_df["MthCalDt"] = pd.to_datetime(rf_stock_selection_df[

rf_monthly_results_df = pd.read_csv("rf_monthly_results_post2022.csv")

print("df_model shape:", df_model.shape)
print("rf_stock_selection_df shape:", rf_stock_selection_df.shape)
print("rf_monthly_results_df shape:", rf_monthly_results_df.shape)

display(rf_stock_selection_df.head(10))
```

```
df_model shape: (1590627, 22)
rf_stock_selection_df shape: (72, 8)
rf_monthly_results_df shape: (12, 6)
```

	rebalance_month	holding_month	PERMNO	MthCalDt	rf_pred	y_next	
0	202412	202501	93436	2024-12-31	0.033954	-0.001818	7.6
1	202412	202501	11308	2024-12-31	0.033954	0.015895	2.4
2	202412	202501	18163	2024-12-31	0.033954	-0.007530	2.24
3	202412	202501	14702	2024-12-31	0.033954	0.105259	2.26
4	202412	202501	38703	2024-12-31	0.033954	0.118168	2.26
5	202412	202501	20357	2024-12-31	0.033954	-0.058285	2.33
6	202501	202502	86580	2025-01-31	0.018532	0.037093	7.0
7	202501	202502	18726	2025-01-31	0.018532	-0.024427	2.59
8	202501	202502	11762	2025-01-31	0.018532	-0.104758	2.38
9	202501	202502	14541	2025-01-31	0.018532	0.071612	2.40

In [2]:

```
# ==========================
# 3. 准备全量月份序列
# 用 cleaned_model_data.csv 的完整历史来找过去36个月
# 这样虽然研究 post-2022，但协方差仍然可以严格取 rebalance month 前36个月
# ==========================
all_months_full = sorted(df_model["DateKey"].unique())

# 只保留 post-2022 里真正出现过的 rebalance months
rebalance_months = sorted(rf_stock_selection_df["rebalance_month"].unique

print("Number of rebalance months:", len(rebalance_months))
print("First 5 rebalance months:", rebalance_months[:5])
print("Last 5 rebalance months :", rebalance_months[-5:])

# ==========================
# 4. 定义一个函数：给定月份，找它之前（含当月）的36个月窗口
# ==========================
def get_rolling_window_months(current_month, all_months, window_size=36):
    idx = all_months.index(current_month)
    if idx < window_size - 1:
        return None
    return all_months[idx - window_size + 1 : idx + 1]

# ==========================
# 5. 定义 Sharpe ratio 最大化问题
# maximize w' mu / sqrt(w' Sigma w)
# scipy 是做最小化，所以返回负的 Sharpe
# ==========================
def negative_sharpe_ratio(weights, mu, cov):
    portfolio_return = np.dot(weights, mu)
```

```
    portfolio_vol = np.sqrt(np.dot(weights.T, np.dot(cov, weights)))

    if portfolio_vol == 0:
        return 1e6

    return -(portfolio_return / portfolio_vol)

# ==========================
# 6. 定义 MVO 求解函数
# long-only, fully invested, optional upper bound
# ==========================
def solve_mvo(mu, cov, upper_bound=0.4):
    n = len(mu)

    # 初始权重：等权
    init_w = np.repeat(1 / n, n)

    # 约束：权重和为1
    constraints = [
        {"type": "eq", "fun": lambda w: np.sum(w) - 1}
    ]

    # 边界：0 <= w <= upper_bound
    bounds = [(0, upper_bound) for _ in range(n)]

    result = minimize(
        fun=negative_sharpe_ratio,
        x0=init_w,
        args=(mu, cov),
        method="SLSQP",
        bounds=bounds,
        constraints=constraints
    )

    if result.success:
        return result.x
    else:
        # 如果优化失败，退回等权
        return init_w
```

```
Number of rebalance months: 12
First 5 rebalance months: [np.int64(202412), np.int64(202501), np.int64(20
2502), np.int64(202503), np.int64(202504)]
Last 5 rebalance months : [np.int64(202507), np.int64(202508), np.int64(20
2509), np.int64(202510), np.int64(202511)]
```

In [4]:

```
# ==========================
# 7. 主循环：对每个 rebalance month 做 MVO
# 基于 RF 已选出的 top 6 股票
# ==========================
mvo_monthly_results = []
mvo_stock_weights = []

for month in rebalance_months:
    # --------------------------
    # A. 取该月 RF 选出的 6 只股票
    # --------------------------
    selected = rf_stock_selection_df[
        rf_stock_selection_df["rebalance_month"] == month
    ].sort_values("rank_in_portfolio").copy()
```

```
    # 必须正好有 6 只
    if len(selected) != 6:
        continue

    stock_ids = selected["PERMNO"].tolist()
    mu = selected["rf_pred"].values              # expected return vector
    realized_next = selected["y_next"].values    # 下一月真实超额收益

    # -------------------------
    # B. 找该月之前（含当月）的过去 36 个月窗口
    # -------------------------
    window_months = get_rolling_window_months(
        current_month=month,
        all_months=all_months_full,
        window_size=36
    )

    if window_months is None:
        continue

    # -------------------------
    # C. 提取这 6 只股票过去 36 个月历史收益
    # 这里用 ExcessRet 来估计协方差矩阵
    # -------------------------
    hist_data = df_model[
        (df_model["DateKey"].isin(window_months)) &
        (df_model["PERMNO"].isin(stock_ids))
    ][["DateKey", "PERMNO", "ExcessRet"]].copy()

    if hist_data.empty:
        continue

    # -------------------------
    # D. 处理重复值：
    # 如果同一 DateKey + PERMNO 有多条记录，先聚合成唯一值
    # 这里取均值
    # -------------------------
    hist_data = (
        hist_data.groupby(["DateKey", "PERMNO"], as_index=False)["ExcessR
        .mean()
    )

    # -------------------------
    # E. 转成宽表：行=月份，列=股票，值=ExcessRet
    # -------------------------
    hist_pivot = hist_data.pivot(
        index="DateKey",
        columns="PERMNO",
        values="ExcessRet"
    )

    # 保证列顺序和 stock_ids 一致
    hist_pivot = hist_pivot.reindex(columns=stock_ids)

    # -------------------------
    # F. 删除含缺失的月份
    # 只保留 6 只股票都有收益的月份
    # -------------------------
    hist_pivot = hist_pivot.dropna()
```

```
    # 至少保留一定数量历史月份，否则协方差不稳定
    if hist_pivot.shape[0] < 12:
        continue

    # -------------------------
    # G. 计算协方差矩阵
    # 为了数值稳定，在对角线上加一个很小的 ridge
    # -------------------------
    cov = hist_pivot.cov().values
    cov = cov + np.eye(len(stock_ids)) * 1e-6

    # -------------------------
    # H. 求解 MVO 最优权重
    # -------------------------
    weights = solve_mvo(mu=mu, cov=cov, upper_bound=0.4)

    # -------------------------
    # I. 计算下一月组合收益
    # MVO 收益 = 权重 × 下一月真实收益
    # 同时保留等权收益做检查
    # -------------------------
    mvo_return = np.dot(weights, realized_next)
    ew_return = np.mean(realized_next)

    # -------------------------
    # J. 保存月度组合结果
    # -------------------------
    mvo_monthly_results.append({
        "rebalance_month": month,
        "holding_month": selected["holding_month"].iloc[0],
        "n_selected": len(stock_ids),
        "n_hist_months_used": hist_pivot.shape[0],
        "rf_mvo_return": mvo_return,
        "rf_ew_return_check": ew_return
    })

    # -------------------------
    # K. 保存每只股票的权重结果
    # -------------------------
    stock_weight_output = selected[
        ["rebalance_month", "holding_month", "PERMNO", "rf_pred", "y_next
    ].copy()

    stock_weight_output["mvo_weight"] = weights
    stock_weight_output["hist_months_used"] = hist_pivot.shape[0]

    mvo_stock_weights.append(stock_weight_output)

print("MVO loop completed.")
```

```
MVO loop completed.
```

In [5]:

```
# ========================
# 8. 拼接结果表
# ========================
mvo_monthly_results_df = pd.DataFrame(mvo_monthly_results)

mvo_stock_weights_df = (
    pd.concat(mvo_stock_weights, ignore_index=True)
```

```
    if mvo_stock_weights else pd.DataFrame()
)

print("mvo_monthly_results_df shape:", mvo_monthly_results_df.shape)
print("mvo_stock_weights_df shape :", mvo_stock_weights_df.shape)

display(mvo_monthly_results_df.head())
display(mvo_stock_weights_df.head(12))

# ========================
# 9. 保存结果
# ========================
mvo_monthly_results_df.to_csv("rf_mvo_monthly_results_post2022.csv", inde
mvo_stock_weights_df.to_csv("rf_mvo_stock_weights_post2022.csv", index=Fa

print("Saved files:")
print("- rf_mvo_monthly_results_post2022.csv")
print("- rf_mvo_stock_weights_post2022.csv")
```

```
mvo_monthly_results_df shape: (11, 6)
mvo_stock_weights_df shape : (66, 8)
```

	rebalance_month	holding_month	n_selected	n_hist_months_used	rf_mvo_retur
0	202501	202502	6	33	-0.00452
1	202502	202503	6	36	-0.06257
2	202503	202504	6	36	0.02376
3	202504	202505	6	36	0.00409
4	202505	202506	6	12	0.02365

	rebalance_month	holding_month	PERMNO	rf_pred	y_next	rank_in_portf
0	202501	202502	86580	0.018532	0.037093	
1	202501	202502	18726	0.018532	-0.024427	
2	202501	202502	11762	0.018532	-0.104758	
3	202501	202502	14541	0.018532	0.071612	
4	202501	202502	14702	0.018532	-0.124846	
5	202501	202502	86783	0.018532	0.055475	
6	202502	202503	86580	-0.029943	-0.135730	
7	202502	202503	78975	-0.029943	-0.003156	
8	202502	202503	57665	-0.029943	-0.200129	
9	202502	202503	86456	-0.029943	-0.039085	
10	202502	202503	75186	-0.029943	-0.019117	
11	202502	202503	13721	-0.029943	-0.001056	

```
Saved files:
- rf_mvo_monthly_results_post2022.csv
- rf_mvo_stock_weights_post2022.csv
```

In [6]:
```
# ==========================
# 10. 计算 MVO 组合表现
# ==========================
portfolio_returns_mvo = mvo_monthly_results_df["rf_mvo_return"].dropna()

annualized_return_mvo = (1 + portfolio_returns_mvo).prod() ** (12 / len(p
annualized_vol_mvo = portfolio_returns_mvo.std() * np.sqrt(12)

if portfolio_returns_mvo.std() != 0:
    sharpe_ratio_mvo = (portfolio_returns_mvo.mean() / portfolio_returns_
else:
    sharpe_ratio_mvo = np.nan

wealth_mvo = (1 + portfolio_returns_mvo).cumprod()
peak_mvo = wealth_mvo.cummax()
drawdown_mvo = (peak_mvo - wealth_mvo) / peak_mvo
max_drawdown_mvo = drawdown_mvo.max()

print("Random Forest + MVO Portfolio Performance (Post-2022 only)")
print("Annualized Return    :", round(annualized_return_mvo, 6))
print("Annualized Volatility:", round(annualized_vol_mvo, 6))
print("Sharpe Ratio         :", round(sharpe_ratio_mvo, 6))
print("Max Drawdown         :", round(max_drawdown_mvo, 6))
```

```
Random Forest + MVO Portfolio Performance (Post-2022 only)
Annualized Return    : 0.074478
Annualized Volatility: 0.113475
Sharpe Ratio         : 0.686859
Max Drawdown         : 0.06257
```

In [7]:
```
# ==========================
# 11. 读取 RF 等权月度结果做对比
# rf_monthly_results_df 是前面已保存的 post-2022 等权结果
# ==========================
rf_monthly_results_df = pd.read_csv("rf_monthly_results_post2022.csv")

portfolio_returns_ew = rf_monthly_results_df["rf_ew_return"].dropna()

annualized_return_ew = (1 + portfolio_returns_ew).prod() ** (12 / len(por
annualized_vol_ew = portfolio_returns_ew.std() * np.sqrt(12)

if portfolio_returns_ew.std() != 0:
    sharpe_ratio_ew = (portfolio_returns_ew.mean() / portfolio_returns_ew
else:
    sharpe_ratio_ew = np.nan

wealth_ew = (1 + portfolio_returns_ew).cumprod()
peak_ew = wealth_ew.cummax()
drawdown_ew = (peak_ew - wealth_ew) / peak_ew
max_drawdown_ew = drawdown_ew.max()

comparison_df = pd.DataFrame({
    "Portfolio": ["RF + Equal Weight", "RF + MVO"],
    "Annualized Return": [annualized_return_ew, annualized_return_mvo],
    "Annualized Volatility": [annualized_vol_ew, annualized_vol_mvo],
    "Sharpe Ratio": [sharpe_ratio_ew, sharpe_ratio_mvo],
    "Max Drawdown": [max_drawdown_ew, max_drawdown_mvo]
})

display(comparison_df.round(6))
```

```
comparison_df.to_csv("rf_portfolio_comparison_post2022.csv", index=False)
print("Saved: rf_portfolio_comparison_post2022.csv")
```

	Portfolio	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
0	RF + Equal Weight	0.210254	0.129452	1.545340	0.08036
1	RF + MVO	0.074478	0.113475	0.686859	0.06257

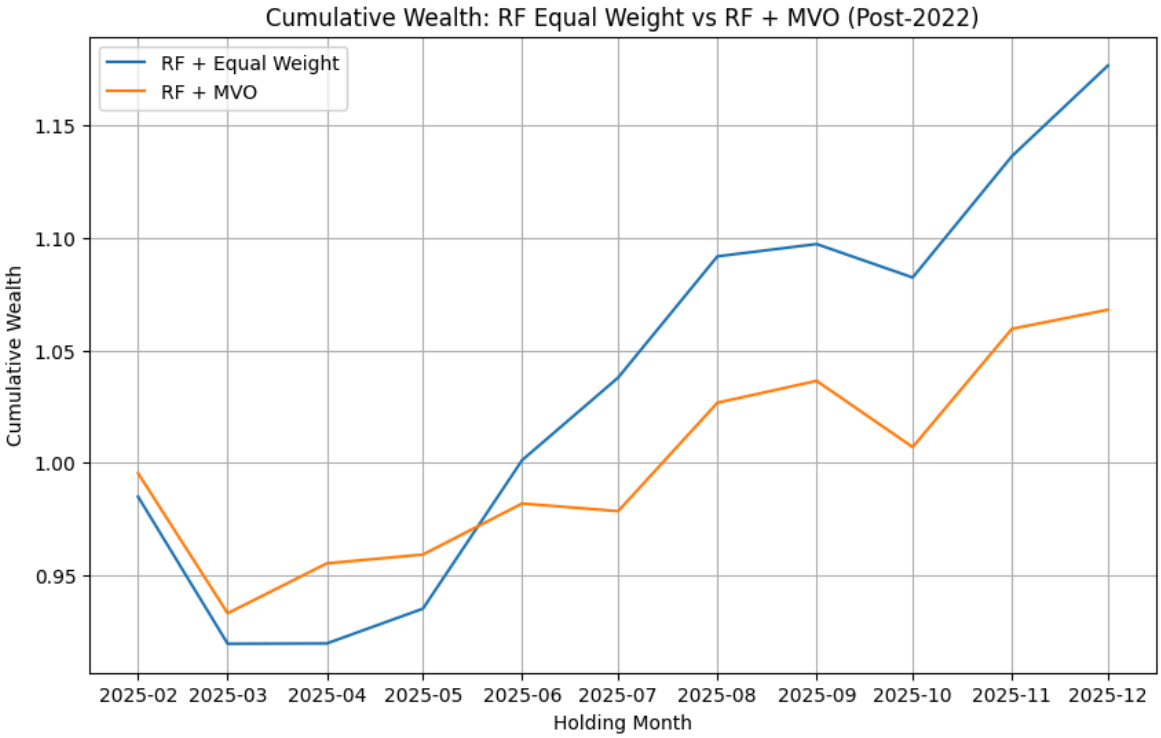
```
Saved: rf_portfolio_comparison_post2022.csv
```

In [8]:

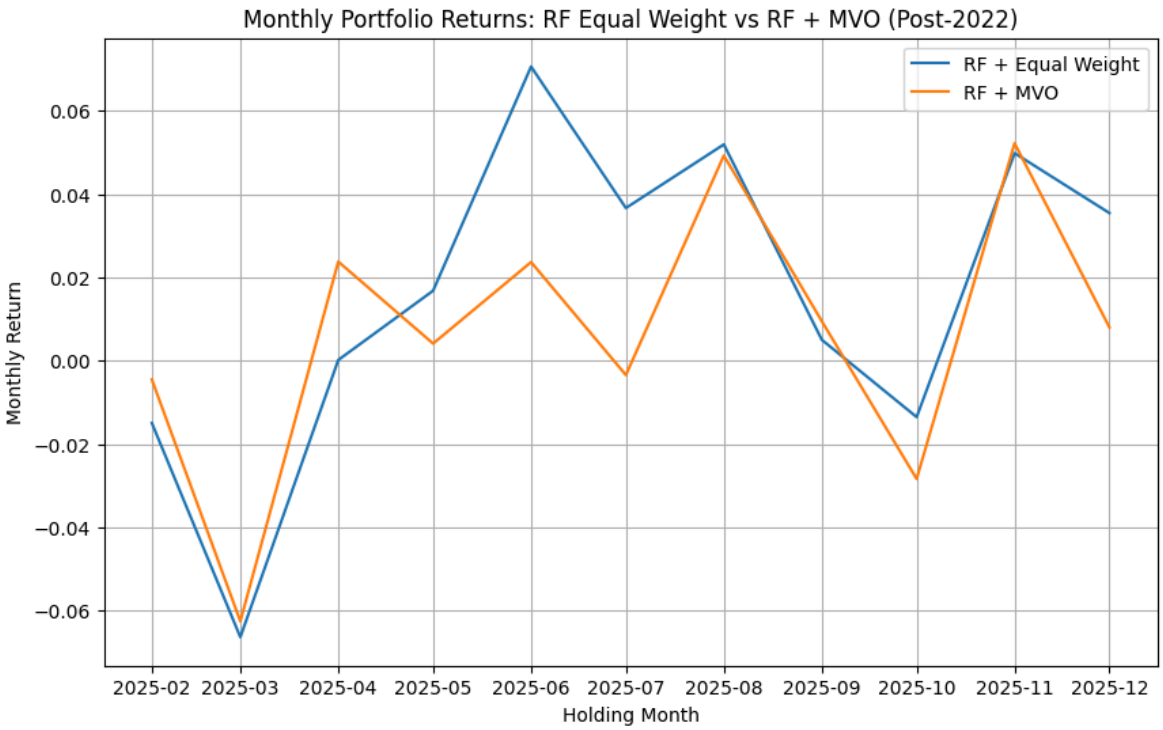
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# ==========================
# 1. 读取结果表
# ==========================
rf_monthly_results_df = pd.read_csv("rf_monthly_results_post2022.csv")
mvo_monthly_results_df = pd.read_csv("rf_mvo_monthly_results_post2022.csv

# ==========================
# 2. 取两种组合的月度收益
# ==========================
ew_returns = rf_monthly_results_df[["holding_month", "rf_ew_return"]].cop
mvo_returns = mvo_monthly_results_df[["holding_month", "rf_mvo_return"]].

# 合并到同一张表
compare_returns = pd.merge(
    ew_returns,
    mvo_returns,
    on="holding_month",
    how="inner"
).sort_values("holding_month").reset_index(drop=True)

# 把 holding_month 转成日期，方便画图
compare_returns["holding_month"] = pd.to_datetime(
    compare_returns["holding_month"].astype(str) + "01",
    format="%Y%m%d"
)

# ==========================
# 3. 计算累计财富
# 初始财富设为 1
# ==========================
compare_returns["wealth_ew"] = (1 + compare_returns["rf_ew_return"]).cump
compare_returns["wealth_mvo"] = (1 + compare_returns["rf_mvo_return"]).cu

display(compare_returns.head())

# ==========================
# 4. 画累计财富曲线
# ==========================
plt.figure(figsize=(10, 6))
plt.plot(compare_returns["holding_month"], compare_returns["wealth_ew"],
plt.plot(compare_returns["holding_month"], compare_returns["wealth_mvo"],

plt.title("Cumulative Wealth: RF Equal Weight vs RF + MVO (Post-2022)")
```

```
plt.xlabel("Holding Month")
plt.ylabel("Cumulative Wealth")
plt.legend()
plt.grid(True)
plt.show()
```

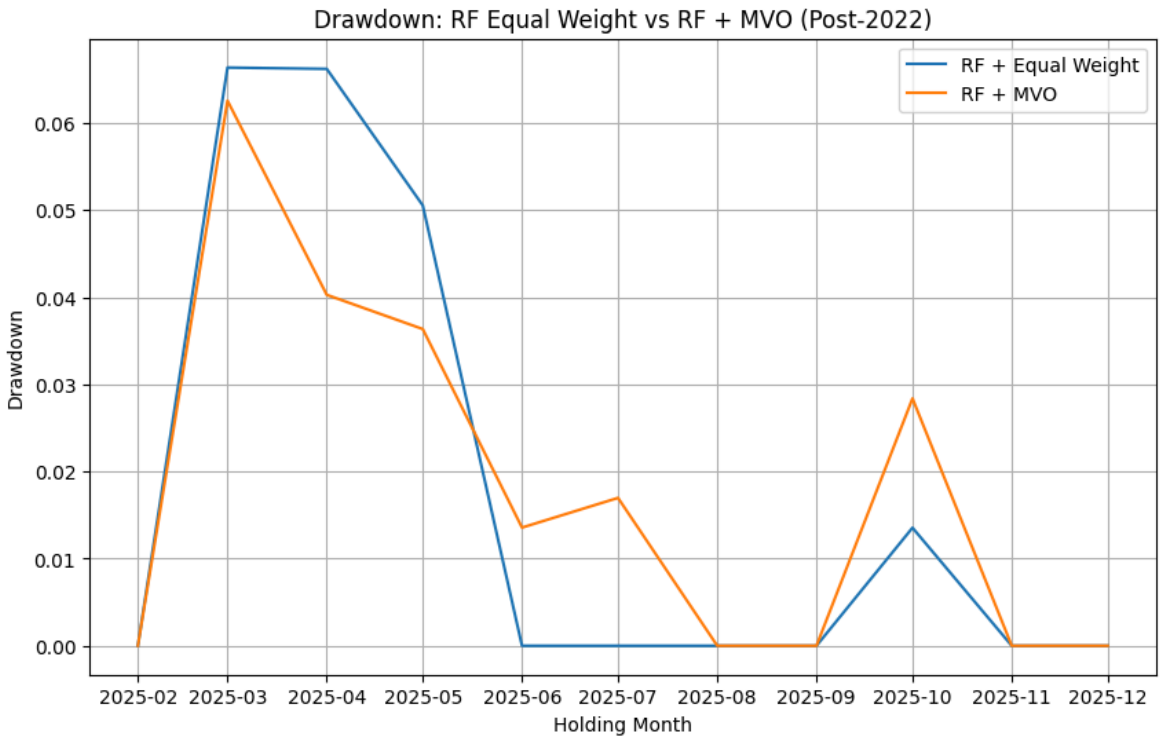
	holding_month	rf_ew_return	rf_mvo_return	wealth_ew	wealth_mvo
0	2025-02-01	-0.014975	-0.004525	0.985025	0.995475
1	2025-03-01	-0.066379	-0.062570	0.919640	0.933188
2	2025-04-01	0.000160	0.023768	0.919788	0.955368
3	2025-05-01	0.016789	0.004095	0.935230	0.959280
4	2025-06-01	0.070591	0.023654	1.001249	0.981971



In [9]:
```
# ========================
# 5. 画月度收益对比图
# ========================
plt.figure(figsize=(10, 6))
plt.plot(compare_returns["holding_month"], compare_returns["rf_ew_return"
plt.plot(compare_returns["holding_month"], compare_returns["rf_mvo_return

plt.title("Monthly Portfolio Returns: RF Equal Weight vs RF + MVO (Post-2
plt.xlabel("Holding Month")
plt.ylabel("Monthly Return")
plt.legend()
plt.grid(True)
plt.show()
```



In [10]:
```
# ==========================
# 6. 计算并画回撤曲线
# ==========================
compare_returns["peak_ew"] = compare_returns["wealth_ew"].cummax()
compare_returns["drawdown_ew"] = (
    compare_returns["peak_ew"] - compare_returns["wealth_ew"]
) / compare_returns["peak_ew"]

compare_returns["peak_mvo"] = compare_returns["wealth_mvo"].cummax()
compare_returns["drawdown_mvo"] = (
    compare_returns["peak_mvo"] - compare_returns["wealth_mvo"]
) / compare_returns["peak_mvo"]

plt.figure(figsize=(10, 6))
plt.plot(compare_returns["holding_month"], compare_returns["drawdown_ew"]
plt.plot(compare_returns["holding_month"], compare_returns["drawdown_mvo"

plt.title("Drawdown: RF Equal Weight vs RF + MVO (Post-2022)")
plt.xlabel("Holding Month")
plt.ylabel("Drawdown")
plt.legend()
plt.grid(True)
plt.show()
```



In [11]:
```
# ==========================
# 12. 每个月 top6 股票 + MVO 权重，整理成宽表
# ==========================
mvo_wide = mvo_stock_weights_df[
    ["rebalance_month", "holding_month", "rank_in_portfolio", "PERMNO", "
].sort_values(["rebalance_month", "rank_in_portfolio"]).copy()

permno_wide = mvo_wide.pivot(
    index=["rebalance_month", "holding_month"],
    columns="rank_in_portfolio",
    values="PERMNO"
)
permno_wide.columns = [f"stock_{i}" for i in permno_wide.columns]

pred_wide = mvo_wide.pivot(
    index=["rebalance_month", "holding_month"],
    columns="rank_in_portfolio",
    values="rf_pred"
)
pred_wide.columns = [f"pred_{i}" for i in pred_wide.columns]

actual_wide = mvo_wide.pivot(
    index=["rebalance_month", "holding_month"],
    columns="rank_in_portfolio",
    values="y_next"
)
actual_wide.columns = [f"actual_{i}" for i in actual_wide.columns]

weight_wide = mvo_wide.pivot(
    index=["rebalance_month", "holding_month"],
    columns="rank_in_portfolio",
    values="mvo_weight"
)
weight_wide.columns = [f"weight_{i}" for i in weight_wide.columns]

mvo_top6_by_month = pd.concat(
    [permno_wide, pred_wide, actual_wide, weight_wide],
```

```
    axis=1
).reset_index()

mvo_top6_by_month.to_csv("rf_mvo_top6_by_month_post2022.csv", index=False

print("Saved: rf_mvo_top6_by_month_post2022.csv")
display(mvo_top6_by_month.head())
```

```
Saved: rf_mvo_top6_by_month_post2022.csv
```

	rebalance_month	holding_month	stock_1	stock_2	stock_3	stock_4	stock_5
0	202501	202502	86580	18726	11762	14541	14702
1	202502	202503	86580	78975	57665	86456	75186
2	202503	202504	84398	66181	65875	66093	89071
3	202504	202505	84398	92027	77178	57665	89469
4	202505	202506	84398	77178	19383	18163	88360